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LIST OF SYMBOLS

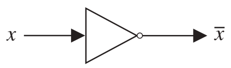
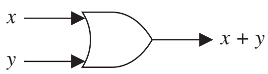
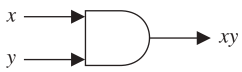
TOPIC	SYMBOL	MEANING	PAGE
LOGIC	$\neg p$	negation of p	3
	$p \wedge q$	conjunction of p and q	4
	$p \vee q$	disjunction of p and q	4
	$p \oplus q$	exclusive or of p and q	5
	$p \rightarrow q$	implication p implies q	6
	$p \leftrightarrow q$	biconditional of p and q	10
	$p \equiv q$	equivalence of p and q	27
	$\mathbf{T}$	tautology	29
	$\mathbf{F}$	contradiction	29
	$P(x_1, \ldots, x_n)$	propositional function	42
	$\forall x P(x)$	universal quantification of $P(x)$	44
	$\exists x P(x)$	existential quantification of $P(x)$	45
	$\exists! x P(x)$	uniqueness quantification of $P(x)$	46
	$\therefore$	therefore	73
	$p\{S\}q$	partial correctness of S	393
SETS	$x \in S$	x is a member of S	122
	$x \notin S$	x is not a member of S	122
	$\{a_1, \ldots, a_n\}$	list of elements of a set	122
	$\{x \mid P(x)\}$	set builder notation	122
	$\mathbf{N}$	set of natural numbers	122
	$\mathbf{Z}$	set of integers	122
	$\mathbf{Z}^+$	set of positive integers	122
	$\mathbf{Q}$	set of rational numbers	122
	$\mathbf{R}$	set of real numbers	122
	$[a, b], (a, b)$	closed, open intervals	123
	$S = T$	set equality	123
	$\emptyset$	empty (or null) set	124
	$S \subseteq T$	S is a subset of T	125
	$S \subset T$	S is a proper subset of T	126
	$\|S\|$	cardinality of S	127
	$\mathcal{P}(S)$	power set of S	128
	$(a_1, \ldots, a_n)$	n-tuple	128
	(a, b)	ordered pair	128
	$A \times B$	Cartesian product of A and B	129
	$A \cup B$	union of A and B	133
	$A \cap B$	intersection of A and B	134
	$A - B$	difference of A and B	135
	$\overline{A}$	complement of A	135
	$\bigcup_{i=1}^{n} A_i$	union of A_i, $i = 1, 2, \ldots, n$	140
	$\bigcap_{i=1}^{n} A_i$	intersection of A_i, $i = 1, 2, \ldots, n$	140
	$A \oplus B$	symmetric difference of A and B	145
	$\aleph_0$	cardinality of a countable set	180
	$\mathfrak{c}$	cardinality of $\mathbf{R}$	185

TOPIC	SYMBOL	MEANING	PAGE
FUNCTIONS	$f(a)$	value of function f at a	147
	$f{:}A \to B$	function from A to B	147
	$f_1 + f_2$	sum of functions f_1 and f_2	149
	$f_1 f_2$	product of functions f_1 and f_2	149
	$f(S)$	image of set S under function f	149
	$\iota_A(s)$	identity function on A	153
	$f^{-1}(x)$	inverse of f	153
	$f \circ g$	composition of f and g	155
	$\lfloor x \rfloor$	floor function of x	157
	$\lceil x \rceil$	ceiling function of x	157
	a_n	term of $\{a_i\}$ with subscript n	165
	$\sum_{i=1}^{n} a_i$	sum of $a_1, a_2, \ldots, a_n$	172
	$\sum_{\alpha \in S} a_\alpha$	sum of a_α over $\alpha \in S$	175
	$\prod_{i=1}^{n} a_n$	product of $a_1, a_2, \ldots, a_n$	179
	$f(x)$ is $O(g(x))$	$f(x)$ is big-O of $g(x)$	217
	$n!$	n factorial	160
	$f(x)$ is $\Omega(g(x))$	$f(x)$ is big-Omega of $g(x)$	227
	$f(x)$ is $\Theta(g(x))$	$f(x)$ is big-Theta of $g(x)$	227
	$\sim$	asymptotic to	231
	$\min(x, y)$	minimum of x and y	281
	$\max(x, y)$	maximum of x and y	282
	$\approx$	approximately equal to	472
INTEGERS	$a \mid b$	a divides b	252
	$a \nmid b$	a does not divide b	252
	a **div** b	quotient when a is divided by b	253
	a **mod** b	remainder when a is divided by b	253
	$a \equiv b \pmod{m}$	a is congruent to b modulo m	254
	$a \not\equiv b \pmod{m}$	a is not congruent to b modulo m	254
	$\mathbf{Z}_m$	integers modulo m	257
	$(a_k a_{k-1} \ldots a_1 a_0)_b$	base b representation	260
	$\gcd(a, b)$	greatest common divisor of a and b	280
	$\operatorname{lcm}(a, b)$	least common multiple of a and b	282
MATRICES	$[a_{ij}]$	matrix with entries a_{ij}	188
	$\mathbf{A} + \mathbf{B}$	matrix sum of **A** and **B**	189
	$\mathbf{AB}$	matrix product of **A** and **B**	189
	$\mathbf{I}_n$	identity matrix of order n	190
	$\mathbf{A}^t$	transpose of **A**	191
	$\mathbf{A} \vee \mathbf{B}$	join of **A** and **B**	192
	$\mathbf{A} \wedge \mathbf{B}$	meet of **A** and **B**	192
	$\mathbf{A} \odot \mathbf{B}$	Boolean product of **A** and **B**	192
	$\mathbf{A}^{[n]}$	nth Boolean power of **A**	193

(List of Symbols continued at back of book)

LIST OF SYMBOLS

TOPIC	*SYMBOL*	*MEANING*	*PAGE*
COUNTING AND PROBABILITY	$P(n, r)$	number of r-permutations of a set with n elements	429
	$C(n, r)$	number of r-combinations of a set with n elements	431
	$\binom{n}{r}$	binomial coefficient n choose r	431
	$C(n; n_1, n_2, \ldots, n_m)$	multinomial coefficient	457
	$p(E)$	probability of E	470
	$p(E \mid F)$	conditional probability of E given F	481
	$E(X)$	expected value of random variable X	503
	$V(X)$	variance of random variable X	513
	C_n	Catalan number	533
	$N(P_{i_1} \ldots P_{i_n})$	number of elements having all the properties $P_{i_j}, j = 1, \ldots, n$	585
	$N(P'_{i_1} \ldots P'_{i_n})$	number of elements having none of the properties $P_{i_j}, j = 1, \ldots, n$	585
	D_n	number of derangements of n objects	589
RELATIONS	$S \circ R$	composite of relations R and S	606
	R^n	nth power of relation R	607
	R^{-1}	inverse relation	609
	s_C	selection operator for condition C	613
	$P_{i_1, i_2, \ldots, i_m}$	projection	614
	$J_p(R, S)$	join	615
	Δ	diagonal relation	628
	R^*	connectivity relation of R	631
	$a \sim b$	a is equivalent to b	639
	$[a]_R$	equivalence class of a with respect to R	641
	$[a]_m$	congruence class modulo m	642
	(S, R)	poset consisting of set S and partial ordering R	650
	$a \prec b$	a is less than b	651
	$a \succ b$	a is greater than b	651
	$a \preccurlyeq b$	a is less than or equal to b	651
	$a \succcurlyeq b$	a is greater than or equal to b	651
GRAPHS AND TREES	(u, v)	directed edge	625
	$G = (V, E)$	graph with vertex set V and edge set E	673
	$\{u, v\}$	undirected edge	674
	$\deg(v)$	degree of vertex v	685
	$\deg^-(v)$	in-degree of vertex v	687
	$\deg^+(v)$	out-degree of vertex v	687
	K_n	complete graph on n vertices	688
	C_n	cycle of size n	688
	W_n	wheel of size n	689
	Q_n	n-cube	689
	$K_{m,n}$	complete bipartite graph of size m, n	691
	$G - e$	subgraph of G with edge e removed	697
	$G + e$	graph produced by adding edge e to graph G	697

TOPIC	*SYMBOL*	*MEANING*	*PAGE*
GRAPHS AND TREES (cont.)	$G_1 \cup G_2$	union of G_1 and G_2	699
	$a, x_1, \ldots, x_{n-1}, b$	path from a to b	714
	$a, x_1, \ldots, x_{n-1}, a$	circuit	714
	$\kappa(G)$	vertex connectivity of G	718
	$\lambda(G)$	edge connectivity of G	720
	r	number of regions of the plane	756
	$\deg(R)$	degree of region R	757
	$\chi(G)$	chromatic number of G	763
	m	greatest number of children of an internal vertex in a rooted tree	784
	n	number of vertices of a rooted tree	788
	i	number of internal vertices of a rooted tree	788
	l	number of leaves of a rooted tree	789
	h	height of a rooted tree	790
BOOLEAN ALGEBRA	$\overline{x}$	complement of Boolean variable x	847
	$x + y$	Boolean sum of x and y	847
	$x \cdot y$ (or xy)	Boolean product of x and y	847
	B	$\{0, 1\}$	848
	F^d	dual of F	852
	$x \mid y$	x *NAND* y	857
	$x \downarrow y$	x *NOR* y	857
		inverter	22, 859
		OR gate	22, 859
		AND gate	22, 859
LANGUAGES AND FINITE-STATE MACHINES	λ	empty string	166, 887
	xy	concatenation of x and y	371
	$l(x)$	length of string x	371
	w^R	reversal of w	380
	(V, T, S, P)	phrase-structure grammar	887
	S	start symbol	887
	$w \rightarrow w_1$	production	887
	$w_1 \Rightarrow w_2$	w_2 is directly derivable from w_1.	887
	$w_1 \overset{*}{\Rightarrow} w_2$	w_2 is derivable from w_1.	887
	$\langle A \rangle ::= \langle B \rangle c \mid d$	Backus–Naur form	893
	(S, I, O, f, g, s_0)	finite-state machine with output	898
	s_0	initial or start state	898
	AB	concatenation of sets A and B	904
	A^*	Kleene closure of A	905
	(S, I, f, s_0, F)	finite-state machine automaton with no output	905
	(S, I, f, s_0)	Turing machine	927